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AI25BTECH11038 – Tejas Uppala

Question

The scalar $\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C}))$ equals
(A) 0
(B) $[\mathbf{ABC}]$
(C) $[\mathbf{ABC}] + [\mathbf{BCA}]$
(D) None of these

Solution

Using the **distributive property** of cross product over addition:

$$((\mathbf{B}+\mathbf{C})\times(\mathbf{A}+\mathbf{B}+\mathbf{C})) = ((\mathbf{B}+\mathbf{C})\times\mathbf{A}) + ((\mathbf{B}+\mathbf{C})\times(\mathbf{B}+\mathbf{C})) \quad (0.1)$$

The cross product of two parallel vectors is zero:

$$((\mathbf{B}+\mathbf{C})\times(\mathbf{A}+\mathbf{B}+\mathbf{C})) = ((\mathbf{B}+\mathbf{C})\times\mathbf{A}) \quad (0.2)$$

Expanding further,

$$((\mathbf{B}+\mathbf{C})\times\mathbf{A}) = (\mathbf{B}\times\mathbf{A}) + (\mathbf{C}\times\mathbf{A}) \quad (0.3)$$

Solution (contd.)

Now,

$$\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = \mathbf{A} \cdot ((\mathbf{B} \times \mathbf{A}) + (\mathbf{C} \times \mathbf{A})) \tag{0.4}$$

$$= (\mathbf{A} \cdot (\mathbf{B} \times \mathbf{A})) + (\mathbf{A} \cdot (\mathbf{C} \times \mathbf{A})) \tag{0.5}$$

Since $\mathbf{A}$ is perpendicular to both $\mathbf{B} \times \mathbf{A}$ and $\mathbf{C} \times \mathbf{A}$, their dot products are zero:

$$\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = 0 \tag{0.6}$$

Final Answer

$$0$$